Higher National Diploma in Engineering

04th Year, First Semester Examination - 2019

CE3221 – Highway and Railway Engineering

Instructions for Candidates:
Answer five questions only

No. of questions: 06
No. of pages: 03
Time: 03 hours

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1).

a) What do you mean by "Transportation System" of a country? (2 marks)

b) "Transportation develops because of several and frequently overlapping factors". Describe five factors to prove the statement. (5 marks)

c) What do you mean by transportation planning? (2 marks)

d) Explain the basic elements of transportation planning. (5 marks)

e) Classify and explain the main types of roads in Sri Lanka according to "Road Development Authority" (4 marks)

f) What are the types of vertical curves? (2 marks)

2).

a) Explain the significance of curves when considering alignment of highways. (4 marks)

b) Draw a simple horizontal curve and mark all elements in the sketch. (4 marks)

c) Suppose the central angle of the curve is 62° 10' and the radius of the curve is 800 m. Calculate the values of tangent length and long chord length. (4 marks)

d) Define the following terms used in highway Engineering.

i. Shoulders

ii. Center median

iii. Cross slope

iv. Side slopes (8marks)

3).

a) What are the stages in highway alignment and detailing? (4 marks)

b) In highway construction, earthwork is a usual and important activity. Describe the importance of earthwork calculation in highway construction. (4 marks)

c) What do you mean by mass haul diagram and list three uses of it? (4 marks)

d) What are the functions of a highway pavement? (4 marks)

e) Highway pavement failure is a major cause which affect the economy of a country. Describe the causes for structural failure of highway pavement. (4 marks)

4).

a) Describe five important properties to be considered on soils which are used in road construction. (5 marks)

b) Describe a laboratory test to determine the wearing resistance of aggregates used in highway construction. (5 marks)

c) Explain the following terms of the soils used in highway construction.

i. Capillarity

ii. Soil elasticity

iii. Soil plasticity

iv. Liquid limit

v. Plastic limit (5 marks)

d) What are the purpose of conducting following tests on bitumen?

i. Softening point test

ii. Penetration test

iii. Ductility test

iv. Flash and fire point test

v. Viscosity test (5 marks)

5).

a) Explain the importance of highway drainage. (5 marks)

b) Sub surface drainage is important factor to enhance the strength of soil. Capillary rise is a one of major factor for reducing the sub surface soil strength. Describe with neat sketch how to control the capillary rise. (5 marks)

c) Assume you are a highway design engineer for a proposed highway construction project. List out the factors that should be considered at the design stage of the project. (5 marks)

d) Patch repairing is an essential item of work to be conducted in maintenance of rigid pavements. Describe the methods of patch repairing when it has shallow and deep depressions. (5 marks)

6).

a) Define the main two types of traffic streams. (4 marks)

b) Define the following terms with neat sketch considering vehicle movements in a traffic stream.

i. Car following

ii. Lane changing

iii. Queuing

iv. Merging

v. Diverging

vi. Weaving (6 marks)

c) Define the following terms used in traffic engineering.

i. Speed

ii. Free flow speed

iii. Journey speed

iv. Model speed

v. Running speed (5 marks)

d) List three basic types of traffic control devices. (3 marks)

e) Briefly discuss the effect of driver behavior cause for road accidents. (2 marks)